# Momo Platform

Group 1 - Assignment 2

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## Team


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# Project Idea

In many cases, the real estate market is only accessible to large investors. Younger people, families etc. have limited financial means to acquire real estate, and if they have, they will only be able to do it with the help of a mortgage, making them dependent on a financial institution. Also their real estate exposure will often be limited to a single property without the ability to diversify across properties or even countries and dominate their overall investments in a way that is unfavourable from the perspective of a healthy, diversified asset allocation.

Additionally, in the home sharing sector the Web2.0 approach to the provision of online services, have led to the centralization and near monopolisation of a once inherently decentralised ecosystem and resulted in problem such as:

- Intermediation
- Exorbitant fees
- Partiality in dispute resolution
- Data opacity

As a result, the trust between hosts (the supply side) and guests (the demand side) as well as towards the corporations and their intermediation platforms themselves has been eroded. Typical families' real estate investments are heavily concentrated in their own homes because they have limited means to diversify their investment ownership into other real estate properties.

To regain lost trust, it's necessary to create a true marketplace for real estate property sharing and investing that brings together hosts and guests, and fractional ownership investors with real estate owners and developers, minimising and decentralising intermediation between them.

Our platform will decentralise both the real estate rental and property ownership market in a unique combination. Our peer-2-peer-property-plattform will enable investors, developers and guests in a flexible and direct way without the need of agencies or financial institutions. The use of blockchain and smart contracts will create the necessary trust, incentives and transparency for so many players to efficiently work together without the need of a middleman. In phase two, Momo Platform will also allow the owners to fractionalise the use of the homes, such as decentralised short- and long-term rentals, the monetization of the real estate will directly be routed to its fractal owners through the use of digital currencies and NFTs.

- ***What problem are you solving and why do you need a token?***

**Phase 1: Property Market**

The platform grants access to the property market which is currently only accessible to big institutional or sophisticated investors to smaller retail investors wishing to access the property market. It would make assets more liquid and fungible.

Currently, real estate is sold as one single property or as a portfolio of properties. Fractional ownership is structured in a time-share agreement for single property or a percentage ownership to the entire portfolio without ownership rights to the actual properties. Investors have little control over which property will be included in the portfolio (funds, trust, etc).

Tokenization allows for a more granular share structure even for a single property. Portfolio in the form of a fund or a trust can be restructured as a range of tokens each with its unique contractual requirements and valuation. This will allow for individual investors to truly diversify their own 'portfolio' based on their preferences on usage, location, and values.

**Phase 2: Rental Market**

Currently centralised entities such as agencies and booking platforms stand between suppliers and customers of short-term rental property.

Our platform does not only cut out the middleman and hence increases the payout to the suppliers and reduces costs for customers, it also gives the community (i.e. token holders) sovereignty over the platform. This means that their data is better protected and they can influence the terms and conditions of the platform.

- ***Why is tokenization the unique solution?***

Platform businesses are a natural monopoly. Once a provider is established, it is very difficult for competitors to enter into the market and the incumbents have only a limited incentive to adjust their behaviour. Centralised property sharing platforms such as Airbnb or HomeShare where hosts and guests are charged a fee for using the platform to connect. Management fees can account for up to 20% of the cost. Airbnb, being the major player, has significant control on the governance, access and fees payable to them as the platform provider. Even if a competitor would enter the market successfully and promise a better governance, this competitor would eventually gain market power but might not necessarily stick to its promises.

Hence in order to reform the broken system, a disruptive solution with a decentralised governance is needed. Building a decentralised, community-owned and governed home sharing network, will facilitate short and long-term stay accommodation discovery, while providing users with greater choice, enabling bookings to be placed using traditional currencies and popular cryptocurrencies. Tokenizing access could bring more users to this space as the payment systems are completely simplified and transaction costs reduced.

Our tokenized solution for real estate ownership will offer:

- Fractionalized real estate assets and reducing the high barrier to entry from large upfront costs making it accessible to smaller investors
- Real property asset diversification and customization to asset classes and revenue streams (e.g. rental or property appreciation)
- Liquidity by increasing the pool of potential investors and providing a secondary market

- Automation and efficiency through smart contracts which can automate paperwork currently provided by intermediaries and centralised exchanges.
- Data transparency of ownership and visible record keeping
- Structured products where other third parties can join our network to offer loans, property insurance and other valuable products

Our tokenized solution for home-sharing will offer:

- Disintermediation
- Significantly lower fees
- Decentralised dispute resolution
- Transparency

By empowering hosts and guests to transact in a peer-to-peer manner and distributing the benefits of this relationship to both parties, a DAO model will restore trust among hosts and guests in their own platform, driving the organic growth of a community-governed travel ecosystem that provides unmatched value for hosts and guests alike.

- ***What kind of token are you choosing?***

The Momo Platform will feature two distinct types of tokens, a fungible platform token (Momo Token) with utility and governance function and a set of NFTs that represent ownership in real estate.

**1. The Momo Token (MOT)**

*Token Utility:*

The Momo Token will be the native token of the platform and shall adhere to the ERC-20 token standard. In doing so, it will be natively supported by a variety of different crypto wallets. The Momo is essentially a utility token that allows for uniform access to the various functions of the platform which are governed by the smart contracts of the platform:

- Users who stake a predefined amount of token get premium access to the platform (additional functionality, no registration and transaction fees)
- Payment of platform registration fees for listing or “tokenizing” properties on the platform;
- Payment of platform transaction fees for buying and selling tokenized properties; and
- Payment of platform transaction fees for listing tokenized properties on the rental listings portion of the platform.

*Payment Functionality:*

The Momo Token as well as other cryptocurrencies and fiat currencies can be incorporated for the payment of:

- the actual consideration for the purchase and sale of tokenized properties;
- for actual rental of properties; and

- essentially any other future value added service that may become available on the platform, i.e.,cleaning services to rental property owners, etc.)

Even though the Momo Token can be used as a means of payment on the platform, we assume that due to the volatility of the value, most people will prefer to transact in more stable coins (e.g. USDC, DAI).

*Governance Functionality:*

The Momo Token will likewise serve as the governance token for the platform. In order to participate, submit proposals, vote on submitted proposals and have a say on decisions that will influence the future of the community, native token holders (including hosts and guests) must assign part of their tokens. Each member's voting power is dependent on the amount of native tokens owned by the member.

*Potential Security Qualification:*

In the beginning (i.e. at token sale) the token will not have any monetary features. Hence, we are of the view that our token should qualify as a utility token in most jurisdictions outside the US. However, it lies in the nature of a governance token that the token holders have the power to change the functionality of the token. Hence, we cannot avoid that there will be proposals to introduce new features to the token that would cause the token to be treated as a security.

**2. Non-Fungible Tokens (NFT) representing real property**

Our platform will give users the ability to invest into specific parcels of lands. Therefore, we need to use a non-fungible token (NFT) to tokenize the real estate. We will use ERC-721 standard to tokenize unique parcels of land.

Cryptographic NFT asset tokens can represent assets that were not easily divisible before, and can now be fractionalized at lower transaction costs than with established centralised systems. Physical goods that are non-divisible can be first tokenized and then divided and sold off in different parts. Fractional tokenized ownership might in certain situations make those assets more liquid and fungible.

These NFT asset tokens can be further fractionalized to allow unique investments tied to a physical object, like unique artwork, real estate, or any other real-world assets and securities. One could tokenize a building, where some tokens could grant simple ownership titles of a fraction of the real estate, while other tokens could grant special privileges like access rights.

The Momo Token, if marketed effectively, can be an integral component of network growth. Specifically, it can be utilised to incentivize engagement of platform users as well as reward them for engaging in certain tasks, etc.[1] Additionally, the Momo Token can be used to donate to industry players and regulators for purposes of incentivizing platform adoption. This could be in the form of donations and airdrops. The initial supply of Momo Token can

[1] Propy Whitepaper at page 28.

be fixed and issued at the time of the initial token sale. This may add a desired deflationary nature to the token.

In the beginning we envision that the parcels will be owned by a trust operated by ourselves ("the Momo Trust"). In the future we will also allow third parties to use their own trusts to mint NFTs on our platform.

- ***What kind of Blockchain do you intend to use?***

**Evaluation Process**

For our project we have evaluated the following Blockchains:

- **Algorand** uses delegated proof of stake and is a carbon neutral blockchain. It is suitable as a sustainable solution to using blockchain.
- **Tezos** (https://wiki.tezosagora.org/learn/baking/proofofstake/consensus) is cost effective.
- **Binance Smart Chain** (alternative to Ethereum 2.0 but centralised) - designed based on Ethereum but with fast transaction (because it is centralised) and is well established.
- **Ethereum 2.0** uses proof of stake and will become deflationary. It is suitable because of its prestige and history.
- **Polkadot** - interoperability but not market-ready
- **Polygon** - faster transaction speed than Ethereum 1.0; functions as a side chain and can layer on top of any blockchain to provide interoperability between different blockchains

We have also considered building our own blockchain for our product, however for cost and security considerations we have decided against it. Also we believe that our project will have to integrate with other decentralised applications in the future, hence, we want to build on a widely accepted standard to make such integration smoother.

**Conclusion**

We are aiming for an Ethereum Virtual Machine (EVM) - compatible blockchain. Since the token might be used as settlement currency within the platform in the future, we aim to use a blockchain with low transaction fees (e.g. Polygon). The platform will leverage both conventional web technology and decentralised blockchain-based Web3 technology to satisfy the requirements of the community. The native token will be implemented according to the ERC-20 Standard. To enable cross-chain interoperability and allow native token holders to use their preferred blockchains, we will also deploy contracts for our native token using analogous standards on other EVM-compatible blockchains and use bridges to allow users to transfer our native token across chains.

- ***Are you leaning towards a more centralised versus decentralised approach?***

By using a decentralised governance mechanism, we are setting the project up in a way that it will become completely decentralised over time. However, in order to kick-start the platform in the beginning we will need a central team, not only for development but also for marketing. We assume that this central team will become less important over time as the platform continues to grow.

Decentralised applications do not look any different from current websites or mobile apps. The front-end represents what you see, and the backend of a decentralised application represents the entire business logic. For the average user, decentralised applications need to look and feel the same as existing applications, which means that they need to be as easy and intuitive to use if they are to be adopted on a larger scale. Currently, wallet software and key management are difficult, which might be a bottleneck to the mass adoption of Web3 applications. Furthermore, wide-scale adoption can only happen if distrust of centralised solutions is high enough to warrant current trade-offs in usability.

Our approach is a home sharing community that chooses to structure and govern itself as a decentralised autonomous organisation (DAO). Our DAO economy will be built around our layer 2 Momo Token, and therefore all Momo Token holders are community members. The DAO structure enables the community to always remain flexible to redefine and reinterpret itself as the community evolves.

- ***How are you thinking of attracting users? (e.g., partnerships, airdrops)***

Target the crypto and already existing Airbnb community promoting the advantages such as:

- Reduced fees - instead of fees as high as 20%
- Give a voice back to the community that will directly influence the direction, operation and governance of the platform.
- Expand payment options - enable various cryptocurrencies to be used for payments that lower the cost of transacting compared to traditional payment methods
- Align interests - instead of being forced into an extraction imperative common with centralised platforms, use a token to align economic interests of users
- Create additional revenue streams - users can earn additional revenue by referring hosts and guests, as well as by participating in user support, community forums and troubleshooting

Starting as a centralised project with key partners of necessary innovation track record and expertise and then use airdrops of utility tokens to early adopters (of different degrees) to move towards decentralisation.

**Loyalty**: The ownership of the native token will encourage community participation by rewarding activities of various types within our community. To ensure that these activities are performed by participants whose interests are aligned with the interest of the wider community, participation will be exclusively for native token holders. Some activities will be onboarding incentives such as joining the community as an Early Bird; and help the community grow by referring others to our community.

**Airdrops**: We will consider airdrops of our Momo Token to other communities that have a high synergy potential with our community.

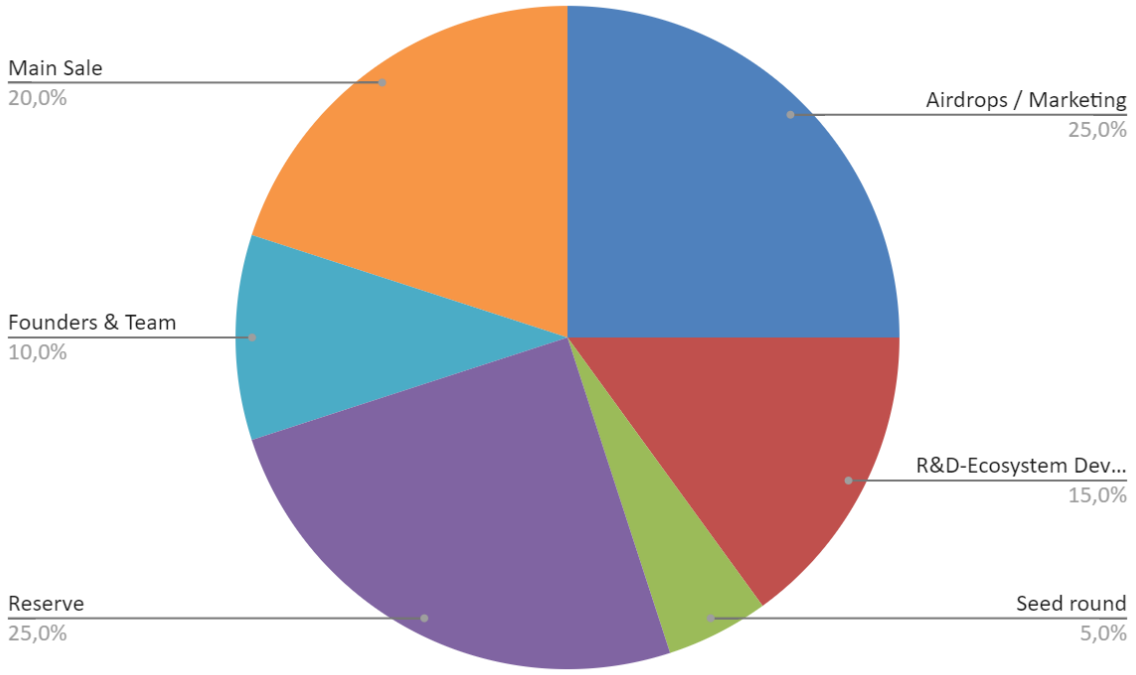
- ***How many tokens are you going to mint?***

On the day of the token release in May 2022, the Momo Token will be launched on the Ethereum blockchain as an ERC-20 token. The maximum total supply will be 1 billion Momo Token. A fixed supply of tokens was chosen for transparency reasons. We are going to set aside a significant token reserve that will allow us to conduct further fundraising rounds if needed without having to dilute existing token holders by minting more tokens.

In theory the total number of tokens is quite arbitrary, since each token can be divided infinitely. The number of 1bn was chosen because it seems to have become a market standard. It also makes sense as a number. It is large enough to allow for a low token price in the beginning (e.g. a fully diluted market cap of $10 Mio. would correspond to a token price of 1 Cent, which can be understood by humans far better than a token price of 0.0001 Cents. Still the token price will be small enough to allow smaller investors to acquire an entire token. Because part of our story is to allow retail customers access to the real estate market, we do not want to discourage small investors by a high price per token (even though they could buy a fraction of a token, it might have a psychological impact.

- ***What is the supply of the token going to look like over time?***

Momo Platform has a fixed supply of 1 billion Momo Tokens. The supply is initially allocated as shown in the following chart.



| Group | % of Tokens | # Tokens | Vesting | Lock up |
|---|---|---|---|---|
| Founders &, Team | 10% | 100’000’000 | 12 month cliff<br>+3 years linear | 12 month cliff<br>+3 years linear |
| R&D, Ecosystem & Advisors | 15% | 150’000’000 | n/a | 3 years linear |
| Seed Round | 5% | 50’000’000 | n/a | 3 years linear |
| Main Sale | 20% | 200’000’000 | n/a | 3 years linear |
| Airdrops / Marketing | 25% | 250’000’000 | n/a | 40 days |
| Reserve | 25% | 250’000’000 | tbd | tbd |
| **Total** | **100%** | **1’000’000’000** | | |

**Founders & Team Members** will be remunerated in Momo Tokens to compensate for their contribution prior to the funding round. After the funding round the team will receive a moderate cash salary and a supplement in token to align their interest with that of the token holders. All team and founder's tokens will be subject to the same vesting conditions, which contains a 12 months cliff vesting followed by a linear vesting over three years. If team members leave the project they will lose all unvested tokens. For team members entering the project the vesting will start at a later date. Those tokens will also be subject to a technical lockup that equals the vesting schedule of the founders. This has no practical impact for the team, but will give assurance to the community that the tokens will not be transferred prior to the vesting. Tokens allocated to **R&D, Ecosystem and Advisors** and other players in the ecosystem will be granted upon completion of a specific task. They will be subject to a 3 years lock-up to avoid sell pressure upon exchange listing. Tokens for **Seed Round** will be subject to a 3 years lock-up to avoid sell pressure upon exchange listing. The tokens reserved for **Airdrops** will be used to incentivise early users (mainly real estate developers) to use the Momo Platform until it has reached a sufficient size to sustain itself. The Airdrops will be subject to a 40 days lockup, for US regulatory reasons.

### Buy-back-and-Burn

Every quarter the Momo Foundation will use a percentage (e.g. 50%) of its incoming revenues from the platform to buy back Momo Tokens from the market and burn them. Through this mechanism we can reduce the number of outstanding Momo Tokens (Total Supply) and hereby increase its value. The Momo Tokenholders will be able to vote on how much of the revenues should be spent on buy-back vs. further development of the platform.

### Staking for Utility

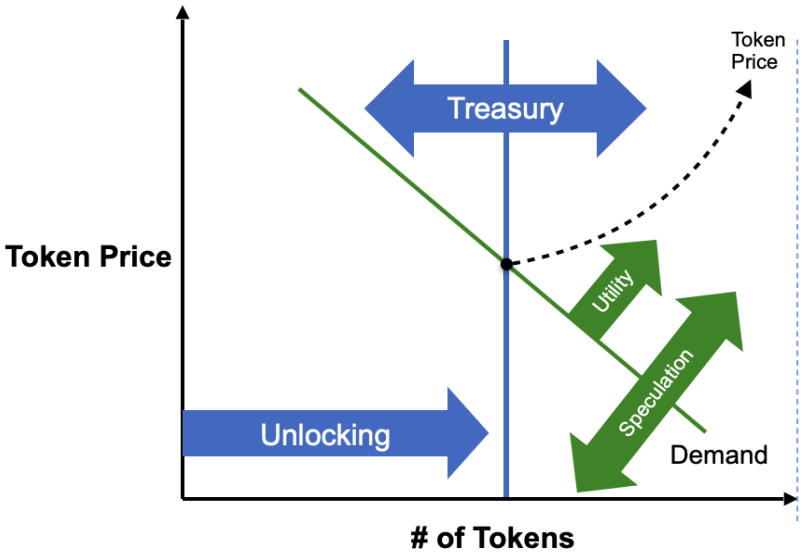
Users will have various ways to stake[2] Momo Tokens in order to get benefits on the Plattform. For example a user who stakes 200’000 Momo Token might get a 20% discount on its transaction fees when selling properties on the Momo Platform. This will generate

[2] Note that the Momo Platform is running on a third-party Layer-1-Blockchain. Therefore it does not have its own consensus mechanisms. “Staking” in the context of the Momo token must therefore not be confused with Staking in the sense of a “Proof-of-Stake”-Blockchain.

additional demand for the token and increase the price of the token as the platform onboards more users.

**Summary**

Overall the forces of supply and demand for the tokens can be summarised in the following chart:




- ***What blockchain are you going to be building off and what specifically is appealing to you about the consensus mechanism that you are using?***

  We will be building on Polygon a EVM - compatible layer 2 solution. This allows us to be compatible with the Ethereum ecosystem and leverage on the security of the Ethereum blockchain, while at the same time benefit from lower gas fees and higher throughput of Polygon. The platform will leverage both conventional web technology and decentralised blockchain-based Web3 technology to satisfy the requirements of the community. The Momo token will be implemented according to the ERC-20 Standard. To enable cross-chain interoperability and allow Momo token holders to use their preferred blockchains, we will also deploy contracts for our native token using analogous standards on other EVM-compatible blockchains and use bridges to allow users to transfer our native token across chains. The Momo Token will adhere to the ERC-20 standard, and our Real-Estate-NFTs to ERC-721. This also allows for the utilisation of a variety of existing wallets which function on the Ethereum blockchain.

- ***How would you go about valuing the token and coming up with a valuation post-launch?***

  Because our project is at seed stage, we attempt to evaluate our utility token with two types of evaluation models for the Pre and Post ICO. Both models are structurally similar to a Discounted Cash Flow model (DCF). Instead of revenues, margins and profits, the Equation of Exchange (MV=PQ) is used to derive each year's current utility value (CUV). Then, since

markets price assets based on future expectations, we discount a future utility value back to the present to derive a rational market price for any given year[3].

**PRE-ICO MODEL**

We consider a general model for valuing our Momo Token Project based on the post "Justified Token Value" written by Indra Adrian Jonklaas posted on Medium web page. This model is based on the Discounted Cash Flow model that is very well established in the world of finance. The assumptions of this model are presented in **Appendix I**.

**POST-ICO MODEL**

For valuing our Momo token, we'll use the Equation of Exchange Model posted by Chris Burniske at Medium. Assumptions and details of this model are in **Appendix II**.

The parameters that we evaluate are:

- M = size of the token base
- V = velocity of the token
- P = price of the digital resource being provisioned
- Q = quantity of the digital resource being provisioned

**Momo Token Value**

As a Pre-ICO valuation, the Present Value (PV) of our project is calculated as $ 111,285,864. Based on a token supply of 1,000,000,000 Momo tokens, each token has a justified value of $0.1 at the point of the project start (the ICO). We understand the market price of a token is based on predictions of the future size of the token economy, and the utility value each token in the float will require to support that economy in the future. According to the Equation of Exchange Model, Momo tokens need to have $0.33 worth of value to support the Momo token economy after the first days of ICO.

| Pre-ICO | Post-ICO | % Increase |
|---|---|---|
| 0.1 | 0.33 | 197% |

- ***What regulatory considerations are you taking into account?***

1. **Jurisdictional Issues**

   Momo Inc is incorporated in the Marshall Islands. Notwithstanding, nodes of the decentralised ledger can and will span locations across the globe creating difficult questions of which jurisdictions' laws and regulations apply to the various aspects of the

[3] Note: We don't use these models as price targets the way equity analysts create price targets for the stocks they cover. We use these models to help to justify our token utility value.

platform. Given the highly regulated nature of real property, it may be appropriate to establish a legal framework and internal governance structure for each jurisdiction in which the platform will offer services.

We intend to market the Momo concept worldwide and in accordance with local laws and regulation. We have decided to not engage in a public offering to potential investors residing in the United States at this time.

Momo Inc intends to comply with local and foreign law. Momo Inc will not be providing exchange based or custodial services. This may change in the future should new regulation be implemented around NFTs creation and a market place for the sale of fractionalized real estate.

## 2. Know Your Customer and Anti-money Laundering Requirements

Together with our legal advisor, we will determine what information must be provided by platform participants in order to use the system. This may involve a tiered approach similar to those adopted by the more centralised crypto exchanges which allow account holders to register for the platform with a bare minimum of disclosed data, i.e., name and email address and then require higher levels of disclosure and documentation depending on which aspects of the platform will be used.

We will investigate KYC/AML integrations with existing and future decentralised self-sovereign identity platforms as well as integrations with the legacy KYC/AML components of state land registry offices.

## 3. Securities Regulations

Our initial round of funding will be a private round. Any US offering will be in accordance with the Securities Act. More specifically and exemption from registration as a security offering will be sought under Rule 506 of, Regulation D. Under Rule 506(c), a company can broadly solicit and generally advertise the offering and still be deemed to be in compliance with the exemption's requirements if:

- The investors in the offering are all accredited investors; and
- The company takes reasonable steps to verify that the investors are accredited investors, which could include reviewing documentation, such as W-2s, tax returns, bank and brokerage statements, credit reports and the like.

Purchasers of securities offered pursuant to Rule 506 receive *"restricted" securities*, meaning that the securities cannot be sold for at least six months or a year without registering them.

Our second round of funding will be for a public offering of the Momo token to be listed on a prominent exchange and subject to the exchanges KYC/AML processes and their obligations.

Beyond initial funding of the legal entity, we will investigate whether the offer of the Momo token or any airdrop promotion itself constitutes security. The Momo token will be used as a means of payment for the registration and development of the NFTs minted for each of the properties. The Momo token will be pegged to a 1 to 1 stablecoin and we intend to develop the Momo through a DAO method and governance over platform development will be through the Momo token holders as described above.

The Momo Token is designed as a utility token for purposes of paying platform fees for services offered. It may be necessary to restrict the use of the Momo to these purposes for fear it may become more than just a utility token.

**4. Privacy Laws**

Momo Inc commits to complying with all relevant local laws in relation to consumer protections, data protection and any privacy rights our customers will have in their countries.

**5. Integration of the Platform with Legacy Land Registry Systems**

At this nascent stage of fractionalising the sale of property, Momo Inc will ensure off-chain processes are also inter-linked with each smart contract. Momo Inc intends to work closely with local land registries in order to further secure the transfer of title ownership to ensure that the real property is not sold a second time off-chain. At the current beta testing stage, a recorded registration of a “Notice to NFT” on title will be sufficient for the purposes of ensuring ownership and rights attached to a property title deed.

**6. Applicable Money Service Business legislation**

Given that the Momo platform will allow users to exchange fiat currency with other tokens for both service charges as well as potential transaction consideration, it will trigger Money Service Business licensing requirements in most jurisdictions. The company intends to apply for such licences in all jurisdictions in which it operates.

# APPENDIX I: Pre-ICO valuation model

**General model assumptions**

To mark model assumptions that drive calculations we use blue font

The model runs quarterly, assuming 36 quarters of “life” (9 years). We have assumed our token will take 9 full years to fully penetrate our Target Market. Additionally, recognizing that scale is becoming easier with modern tech stacks and the influx of Venture Capital and ICO funds, 9 years seems like a reasonable length of time for a platform to scale to the point of capturing the vast majority of the market.

- Cash flow beyond quarter 36 are accounted for by a Terminal Value
- User growth driving transaction growth over time
- Flow through of fees to token holders (the token holder cash flow)
- We use a Terminal Value to account for fees beyond the time period explicitly modelled
- We value the fees to token holders based on a Discount Rate
- Momo Token supply: fixed at 1,000,000,000 tokens
- We suppose a maximum number of users: 100,000,000
- Momo token platform fees: 1% of transaction value; the token holders collectively receive 100% of platform fees.

The ecosystem growth like S-curve distribution, modelled using a logistic function . We consider the S-curve distribution more appropriate for modelling the user adoption and growth, because it models more correctly the slow build up of users as the initial users of the technology; then the innovators are followed by early adopters; and accelerating user growth as the early majority discover the platform; and slowing growth as the late majority and laggards finally adopt the tech many years later.

**Discounted Cash Flow Assumptions**

Recap: Users transact on the Momo token platform. The platform charges 1% of transaction volume as fees; token holders are entitled to a share of the fees (100%), pro-rata based on the proportion of tokens they own relative to total token supply.

We assume 100% of users transact; and the average transaction value will be equal to USD$ 100, a slightly cheaper than the average Airbnb book (USD$ 185). Because the platform charges 1% fees, 100% fees flow through to users; hence for every $100 of transactions on the platform, the holders of tokens earn $1 in distributions.

The maximum distribution to token holders per quarter is $ 100,000,000 when the Target Market is fully penetrated.

We assumed all transactions per user are net of cost (not cost assumption).

Distribution to all tokens holders starts at $20,343 for Q1 2022. At the inflection point, distributions are $62,245,933 at Q3 2026. By Q1 2029 approximately more than 90% of the platform's total distribution per quarter is achieved. By Q2 2030, 99% is achieved.

The Terminal Value of distribution to token holders into the distant future can be modelled as a perpetuity using the formula:

*Terminal Value= (Quarter 36 distribution)／(Discount Rate)*

These Cash Flow to token holders including the Terminal Value are discounted back to the present using a Quarterly Discount Rate of 12.5%, equivalent to an annual Discount Rate of 60%. We decided to use this Discount Rate according to the Probability of Success and Stage of the project, presented by Jonklaas in his post. Because we suppose that our project is in a Seed Stage, and we have faith in it, we'll use the lower bound of the Seed Stage range:

| **Stage** | **Discount Rate** | **Criteria** |
|---|---|---|
| Seed | 60-80% | Size of the opportunity; team; MVP (Minimum Viable Product) |
| Round 1 | 40-50% | Marketing and Sales KPIs: MRR, DAU, MAU, LTV, CAC, CRR, etc. |
| Round 2 | 20-30% | |
| Round 3 | 17.5-22.5% | Financial ratios; Relative to competitors |

**S-curve**

The S-curve is described by the logistic function: $Y=L/((1+e^{(-k(X-X_0))}))$

Where in our project:

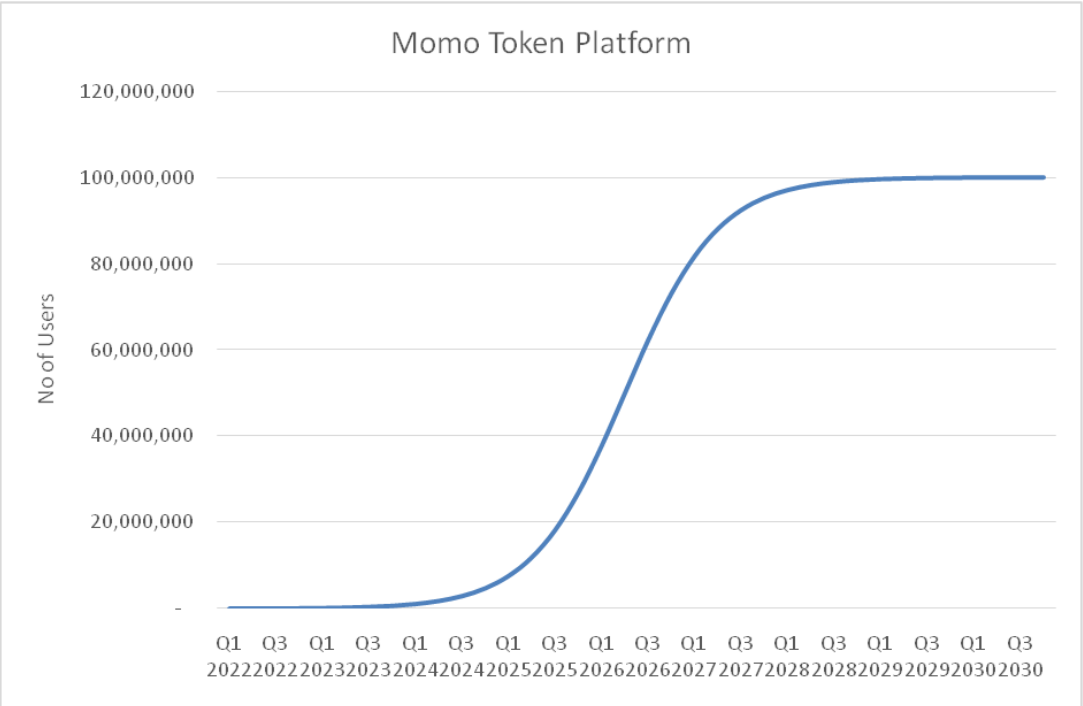
L is the total number of users on the Momo platform

k is a constant that dictates the steepness of the S-curve around the inflection point $X_0$

$X_0$ is the inflection point of the S-curve, which is typically half of the maximum X

X represents time, in this case quarters

In this first part of the model, the key variables are L and k. Because a k value greater than 1 leads to a very steep curve around $X_0$, we discard this value. k-values below 0.3 lead to abnormally high starting user numbers. For this reason, we decide to use 0.5 as a middle value for k. The Momo's S-curve is the following:



# APPENDIX II: Post-ICO valuation model

**Assumptions Equation of Exchange Model:**

**MV = PQ**

**M** = Size of the Token Base. We use our vesting table to determine how many tokens will be in the float and therefore available to the Momo economy. At the ICO moment, 25% of the total that will ever be in existence, and we suppose there will be instant liquidity for those tokens. The remaining tokens are locked up by early private investors, the foundation and the founders, vesting over different periods. We consider another technical aspect like the amount of bonding tokens necessary by the nodes that support our network. Such a bond incentivizes these nodes to act properly. Beyond these bonders, there's the percentage being hodl'd by those that expect the token`s purchasing power to go up, and therefore use it as a store of value to cash in on at a future date.

**V** = Velocity of the token. To calculate the velocity, we use a hybrid velocity, that considers the percentage used as investment and as transactional medium, and an estimation of the number of transactions done with our token. The Hybrid Velocity formula is

*Hybrid Velocity = (%Token for Use 1) x (V1) + (%Token for Use 2) x (V2)*

Where:

*Hybrid Velocity*: Estimation of the number of transactions. We estimate 6 transactions by user.

*Use 1*: Strictly as an Investment. We use the percentage held by Investors at the ICO's beginning.

V*1:* Because these tokens are for investment purpose, we assume velocity is zero (0)

*Use 2:* Transactional Medium. We use the percentage used for utility at the ICO's beginning.

*V2:* Variable to resolve.

Solving to V2, we get a velocity of 14.

**P** = Price of the digital resource being provisioned. We use the cost per GB for user of Momo VPN network. To calculations we use data for:

How Much Does Data Really Cost an ISP? - BroadbandNow

Cisco Annual Internet Report - Cisco

**Q** = According to Savills.com, the real estate market is a $325 tn industry. Real estate is more valuable than all global equities and debt securities combined and almost four times that of global GDP. The current market cap of publicly traded REITs is $2 tn (source Greenstreet.com), and the market cap of tokenized real estate assets as of February 2020

was less than $30m (source stomarket.com). The market potential for Momo tokens is considerable given that only 0.01% of the addressable market has been tokenized, and this market is growing over 500% per year (source reit.com). To make projections about Momo penetration, we use a S-curve, described in Appendix I.

As a discount rate, we keep the 60%/yr as in the Pre-ICO model. Based on all these assumptions, we get a valuation of $0.33 by token.

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